

Installation SIPLACE Connector

Subjects/Contents

1	Objectives.....	3
2	Installation UPD for the Siplace domaine.....	3
3	Workplace/machine configuration HYDRA	3
3.1	Workplace/machine	3
3.1.1	Configuration	3
3.1.2	Machine status	4
3.2	Material buffer and type	7
4	Basic configuration for a Siplace connector.....	8
4.1	Connecting data "Connector" to HYDRA.....	8
4.2	Configuration of lines and machines	10
4.3	Assignment of status.....	13
4.4	Transfer of batch data and update	15
5	Siplace configuration.....	16
5.1	Traceability mode	17
5.2	MES order number	17
6	Available function tests	19
6.1	Transfer of batch data list to the SetupCenter of Siplace	19
6.2	Machine status.....	20
6.3	Order logon / interruption	21
6.4	Create output batches.....	21
6.5	Traceability data	21

1 Objectives

This document describes which configurations are required to successfully connect HYDRA to a Siplace machine using a Siplace connector.

2 Installation UPD for the Siplace domaine

The update package must be installed using the webservice of the Siplace domain for a subsequent installation. Please refer to the installation guide.

3 Workplace/machine configuration HYDRA

3.1 Workplace/machine

A separate workplace/machine must be created in HYDRA for each Siplace line to transfer data to HYDRA.

3.1.1 Configuration

Request menu	Masterdata-> Workplace/machine-> Workplace configuration
Transaction code	res

In general, customary regulations/instructions apply to configure workplaces/machines.

The following requirements are mandatory.

Main tab "Resource configuration"

"Workplace configuration" tab

Workplace type -> N (machine)

Workplace category -> E (individual workplace)

Mandatory sequence -> N (deactivated)

Several logged OPs -> N (only one OP can be logged)

Quantities tab

Manual entry of yield -> yes

Posting of yield as cycles -> yes

MDE configuration tab

Type of monitoring -> no monitoring

3.1.2 Machine status

Request menu	Master data -> workplace/machine -> assigning status
Transaction code	mst

In the status assignment the tab for the status change "Manually at the terminal" must be activated for each status at the Siplace machine

The following statuses are supplied in the standard configuration by the Sidplace adapter.
These statuses must be assigned to HYDRA according to the Sidplace machine.

Status ID	Status text	Comment
1	Production	Sidplace usage "Active"
2	Waiting	
3	Blocked	
4	Interrupted	
5	Malfunction	
6	Setup	
7	Maintenance	
99	General disturbance	
30000	Not assigned	

Other status numbers/statuses can be defined if one or several status numbers are assigned. Please customize the configuration in the Sidplace connector accordingly.

-> File for Sidplace connector config.xml

Example:

```
<Parameter Name="[Laufend]" Value="1"/>
```

<Parameter Name="[Wartend]" Value="2"/>

<Parameter Name="[Blockiert]" Value="3"/>

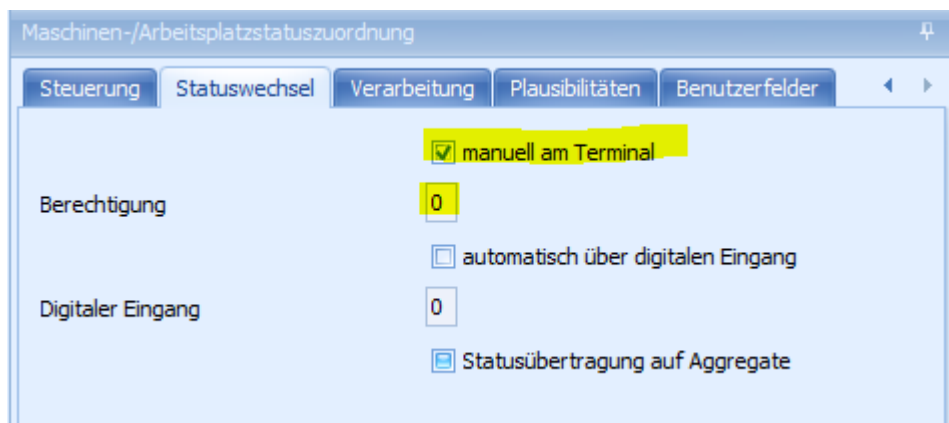
<Parameter Name="[Unterbrochen]" Value="4"/>

<Parameter Name="[Störung]" Value="5"/>

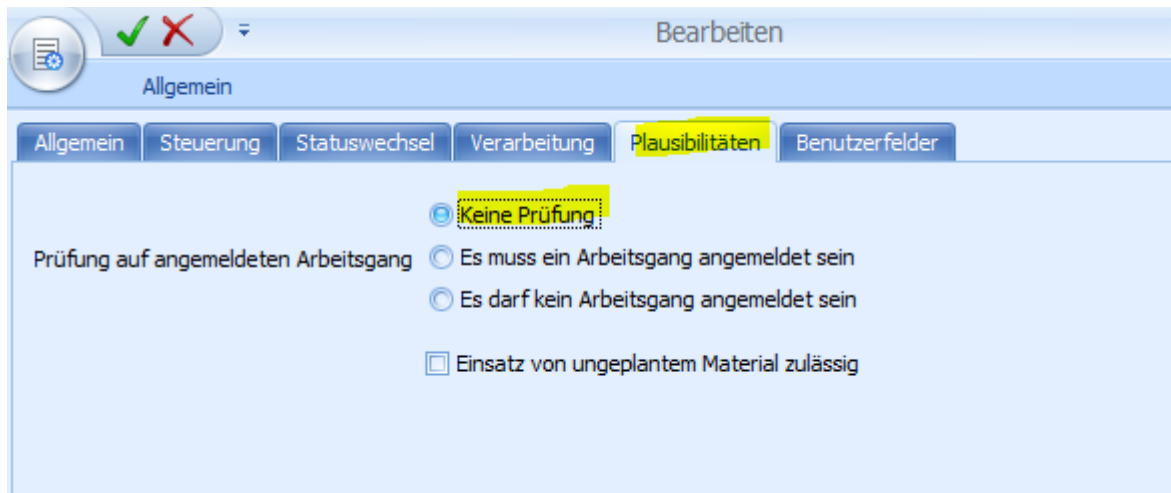
<Parameter Name="[Rüsten]" Value="6"/>

<Parameter Name="[Wartung]" Value="7"/>

<Parameter Name="[NichtZugeordnet]" Value="30000"/>



Also, you must select in the tab "Plausibilities" the option "No check".



3.2 Material buffer and type

In order to transfer input batches to the Sidplace line, you must define a material buffer (or several) in HYDRA. All batches located in this material buffer with a defined material type are transferred into the SiplaceSetup Center.

Configuration in the Sidplace connector

```
-<ComponentConfiguration Name="BatchListGetter">
```

```
<Parameter Name="MaterialType" Value="AIRBAG"/>
```

```
<Parameter Name="MaterialBuffer" Value="MP_V_50510,MP_N_50500"/>
```

```
</ComponentConfiguration>
```

4 Basic configuration for a Siplace connector

4.1 Connecting data "Connector" to HYDRA

In order for the machine data collector to exchange data to and from HYDRA and to receive data from the OIB, you must maintain the connecting parameter in the file config.xml.

The file config.xml is stored in the directory of the Siplace connector.

You must store connecting data to the HYDRA server in the file config.xml of the following section:

```
<ComponentConfiguration Name="WebServiceConnector"
Class="mpdv.MachineDataCollector.ConnectorPlugin.Hydra.WebServiceConnector">

    <Parameter Name="Host" Value="<IP-Adresse HYDRA Leitrechner>" />

    <Parameter Name="Port" Value="<Port>" />

    <Parameter Name="User" Value="<HYDRA User>" />

    <Parameter Name="Pwd" Value="<Passwort HYDRA User>" />

</ComponentConfiguration>
```

Example:

```
<ComponentConfiguration Name="WebServiceConnector"
Class="mpdv.MachineDataCollector.ConnectorPlugin.Hydra.WebServiceConnector">

    <Parameter Name="Host" Value="192.168.94.36" />
```

```
<Parameter Name="Port" Value="8080" />

<Parameter Name="User" Value="12345" />

<Parameter Name="Pwd" Value="mpdv" />

</ComponentConfiguration>
```

You must store connecting data to the HYDRA server in this section of the file config.xml.

```
<ComponentConfiguration Name="SiplaceMonitoringConnector"
Class="mpdv.MachineDataCollector.ConnectorPlugin.Siplace.SiplaceMonitoringConnector">
    <Parameter Name="CallbackEndpoint" Value="http://<IP-Adresse
OIB>.4444/MachineDataCollector.MonitoringEndpoint" />
    <Parameter Name="SubscriptionManagerEndpoint" Value="http:// <IP-Adresse
OIB>.1405/Asm.As.Oib.WS.Eventing.Services/SubscriptionManager" />
</ComponentConfiguration>

<ComponentConfiguration Name="SetupCenterConnector"
Class="mpdv.MachineDataCollector.ConnectorPlugin.Siplace.SetupCenterConnector">
    <Parameter Name="SetupCenterEndpoint" Value="http:// <IP-Adresse
OIB>.1405/Asm.As.Oib.SetupCenter/SiplaceSetupCenter" />
</ComponentConfiguration>

<ComponentConfiguration Name="SiplaceTracingConnector"
Class="mpdv.MachineDataCollector.ConnectorPlugin.Siplace.SiplaceTracingConnector">
    <Parameter Name="SubscriptionManagerEndpoint" Value="http:// <IP-Adresse
OIB>.1405/Asm.As.Oib.WS.Eventing.Services/SubscriptionManager" />
</ComponentConfiguration>
```

Example:

```
<ComponentConfiguration Name="SiplaceMonitoringConnector"
Class="mpdv.MachineDataCollector.ConnectorPlugin.Siplace.SiplaceMonitoringConnector">
  <Parameter Name="CallbackEndpoint"
Value="http://192.168.62.20:4444/MachineDataCollector.MonitoringEndpoint" />
  <Parameter Name="SubscriptionManagerEndpoint"
Value="http://192.168.62.20:1405/Asm.As.Oib.WS.Eventing.Services/SubscriptionManager" />
</ComponentConfiguration>

<ComponentConfiguration Name="SetupCenterConnector"
Class="mpdv.MachineDataCollector.ConnectorPlugin.Siplace.SetupCenterConnector">
  <Parameter Name="SetupCenterEndpoint"
Value="http://192.168.62.20:1405/Asm.As.Oib.SetupCenter/SiplaceSetupCenter" />
</ComponentConfiguration>

<ComponentConfiguration Name="SiplaceTracingConnector"
Class="mpdv.MachineDataCollector.ConnectorPlugin.Siplace.SiplaceTracingConnector">
  <Parameter Name="SubscriptionManagerEndpoint"
Value="http://192.168.62.20:1405/Asm.As.Oib.WS.Eventing.Services/SubscriptionManager" />
</ComponentConfiguration>
```

4.2 Configuration of lines and machines

HYDRA displays each assembly line as a machine of the resource type "MNR". You must store essential data in the file config.xml of the connector in order for it to display messages from stations and lines on "HYDRA" machines.

To do so, you must carry out a component configuration for each assembly line. The configuration contains:

Name of the line (within the Siplace software)

```
<Parameter Name="LineName" Value="Linie_1" />
```

Stations assigned to the line

```
<Parameter Name="StationList" Value="L1_SX2,L1_X3,L1_SX4,L1_X4" />
```

```
<ComponentConfiguration Name="LineStateMerger1"
Class="mpdv.MachineDataCollector.Core.Components.LineStateMerger">
  <Parameter Name="StatePriority" Value="30000,5,4,6,7,3,2,1" />
  <Parameter Name="LineName" Value="Linie_1" />
  <Parameter Name="StationList" Value="L1_SX2,L1_X3,L1_SX4,L1_X4" />
  <Parameter Name="SuppressUnknownStations" Value="true" />
</ComponentConfiguration>
```

Please note:

The parameter "StatePriority" and "SuppressUnknownStations" are part of the configuration to process statuses.

Please set SuppressUnknownStation with the value=true. This ensures that unknown status messages are suppressed and not processed.

The parameter "StatePriority" controls the weighting of statuses in order to generate a status for all machines. This status derives from individual stations. The procedure is as follows: Data is displayed in ascending order by weighting. The first status number stated has the lowest weighting and the last one the highest.

The statuses for all machines is a weighting of the status messages from individual stations. The status of the station with the highest weighting becomes the status of all machines.

```
<Parameter Name="StatePriority" Value="30000,5,4,6,7,3,2,1" />
```

```
<Parameter Name="SuppressUnknownStations" Value="true" />
```

Example:

Example of configuration of two lines (Linie_1 and Linie_2):

```
<ComponentConfiguration Name="LineStateMerger1"
Class="mpdv.MachineDataCollector.Core.Components.LineStateMerger">
  <Parameter Name="StatePriority" Value="30000,5,4,6,7,3,2,1" />
  <Parameter Name="LineName" Value="Linie_1" />
  <Parameter Name="StationList" Value="L1_SX2,L1_X3,L1_SX4,L1_X4" />
  <Parameter Name="SuppressUnknownStations" Value="true" />
</ComponentConfiguration>

<ComponentConfiguration Name="LineStateMerger2"
Class="mpdv.MachineDataCollector.Core.Components.LineStateMerger">
  <Parameter Name="StatePriority" Value="30000,5,4,6,7,3,2,1" />
  <Parameter Name="LineName" Value="Linie_2" />
  <Parameter Name="StationList" Value="L2_S27_1,L2_S27_2,L2_SX2_3,L2_SX2_4" />
  <Parameter Name="SuppressUnknownStations" Value="true" />
</ComponentConfiguration>
```

Assignment of lines for the machines created in HYDRA is done in the following section:

```
<ComponentConfiguration Name="MachineNameTranslator"
Class="mpdv.MachineDataCollector.Core.Components.ParameterTranslator">
  <Parameter Name="<Linienname>" Value="<Maschine>" />
</ComponentConfiguration>
```

Note: <Line name> refers to the names of the lines in the Siplace software

Example:

Assignment of Linie_1 to HYDRA machine SIPLA1

Assignment of Linie_2 to HYDRA maschine SIPLA2

```
<ComponentConfiguration Name="MachineNameTranslator"
Class="mpdv.MachineDataCollector.Core.Components.ParameterTranslator">
  <Parameter Name="[Linie_1]" Value="SIPLA1" />
  <Parameter Name="[Linie_2]" Value="SIPLA2" />
</ComponentConfiguration>
```

4.3 Assignment of status

Siplace reports several statuses. These statuses are assigned to a configured HYDRA machine status. Several Siplace statuses can be assigned to one HYDRA status.

Example:

```
<Parameter Name="[Stop]" Value="interrupted" />
```

Explanation: [Stop] is the status of the Siplace line and „Interrupt“ is the assigned status from HYDRA.

Also, a status number is assigned to each HYDRA status. Status names and numbers must be identical to the machine status created in HYDRA.

Example:

```
<Parameter Name="[Laufend]" Value="1" />
```

Explanation: [Laufend] is the configured status in HYDRA , and „1“ the status number from HYDRA

If no Siplace status is assigned to a HYDRA status, then "-" must be entered into the corresponding field value.

Example:

```
<Parameter Name="[Holiday]" Value="-" />
```

Explanation: [Holiday] is the status of the Siplace line and by entering „-“ you can ensure that the status is not displayed on an existent HYDRA status. Therefore this status is being ignored.

The assignment is file config.xml in the following section configured:

```
<ComponentConfiguration Name="MachineStatusTranslator"  
  Class="mpdv.MachineDataCollector.ConnectorPlugin.Siplace.MachineStatusTranslator">
```

```
<Parameter Name="[Processing]" Value="Laufend" />
<Parameter Name="[Invalid]" Value="NichtZugeordnet" />
<Parameter Name="[PCBBegin]" Value="Laufend" />
<Parameter Name="[PCBBegin2]" Value="Laufend" />
<Parameter Name="[PCBEnd]" Value="Laufend" />
<Parameter Name="[PCBEnd2]" Value="Laufend" />
<Parameter Name="[BreakBegin]" Value="Unterbrochen" />
<Parameter Name="[EmergencyStop]" Value="Unterbrochen" />
<Parameter Name="[Air]" Value="Störung" />
<Parameter Name="[Stop]" Value="Unterbrochen" />
<Parameter Name="[WaitPCBin]" Value="Wartend" />
<Parameter Name="[WaitPCBInside]" Value="Laufend" />
<Parameter Name="[WaitPCBOut]" Value="Blockiert" />
<Parameter Name="[WaitData]" Value="Wartend" />
<Parameter Name="[FiducialError]" Value="Störung" />
<Parameter Name="[TrackError]" Value="Störung" />
<Parameter Name="[MachineError]" Value="Störung" />
<Parameter Name="[TransportError]" Value="Störung" />
<Parameter Name="[BarcodeError]" Value="Störung" />
<Parameter Name="[HeadStep]" Value="Unterbrochen" />
<Parameter Name="[KeySlow]" Value="Unterbrochen" />
<Parameter Name="[Vision]" Value="-" />
<Parameter Name="[Function]" Value="-" />
<Parameter Name="[Init]" Value="Rüsten" />
<Parameter Name="[Holiday]" Value="-" />
<Parameter Name="[Maintenance]" Value="Wartung" />
<Parameter Name="[Setup]" Value="Rüsten" />
<Parameter Name="[DownPlan]" Value="Wartung" />
<Parameter Name="[DownIll]" Value="Störung" />
<Parameter Name="[Prototype]" Value="Rüsten" />
<Parameter Name="[StandAloneBegin]" Value="-" />
<Parameter Name="[StandAloneEnd]" Value="-" />
<Parameter Name="[Rx1]" Value="-" />
<Parameter Name="[Rx2]" Value="-" />
<Parameter Name="[Rx3]" Value="-" />
```

```
<Parameter Name="[FunctionBegin]" Value="-" />
<Parameter Name="[FunctionEnd]" Value="-" />
<Parameter Name="[ErrorEnd]" Value="-" />
<Parameter Name="[BreakEnd]" Value="-" />
<Parameter Name="[Placing]" Value="-" />
<Parameter Name="[Start]" Value="-" />
<Parameter Name="[Laufend]" Value="1" />
<Parameter Name="[Wartend]" Value="2" />
<Parameter Name="[Blockiert]" Value="3" />
<Parameter Name="[Unterbrochen]" Value="4" />
<Parameter Name="[Störung]" Value="5" />
<Parameter Name="[Rüsten]" Value="6" />
<Parameter Name="[Wartung]" Value="7" />
<Parameter Name="[NichtZugeordnet]" Value="30000" />
</ComponentConfiguration>
```

4.4 Transfer of batch data and update

Please carry out a backup of the Siplace database before transferring the batch data list. Also, back up data of the SetupCenter where batch data is managed.

Timing of batch data transfer and updating batch quantities in HYDRA

Batch data transfer to Siplace is set in default with a time interval of 10 seconds. (Transfer of HYDRA to Siplace / SetupCenter)

```
<Parameter Name="[GetBatchList]" Value="10"/>
```

Batch data is updated in a time interval of 20 seconds in HYDRA (transfer of current batch quantities from Sidplace to HYDRA).

```
<Parameter Name="[UpdateBatchConsumption]" Value="20"/>
```

You must adapt the periods depending on the requirement and data quantity.

You can configure the batch data transfer in the file conf.xml in the following section:

```
<ComponentConfiguration Name="TimedTrigger"
Class="mpdv.MachineDataCollector.Core.Components.TimedTrigger">

    <Parameter Name="[GetBatchList]" Value="<in seconds>" " />

    <Parameter Name="[UpdateBatchConsumption]" Value="<in seconds>" />

</ComponentConfiguration>
```

Example:

```
-<ComponentConfiguration Name="TimedTrigger">

    <Parameter Name="[GetBatchList]" Value="10"/>

    <Parameter Name="[UpdateBatchConsumption]" Value="20"/>

</ComponentConfiguration>
```

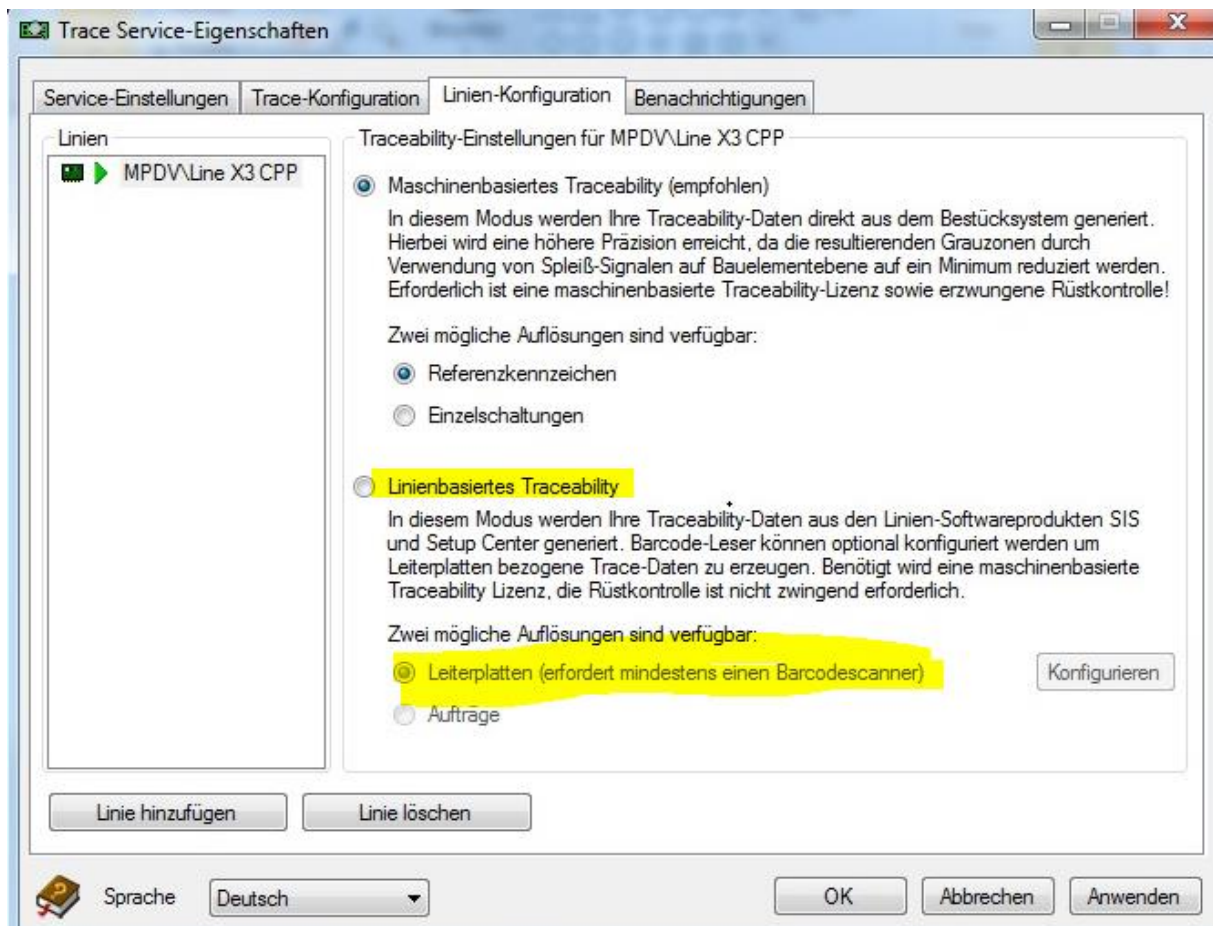
5 Siplace configuration

Requirements are:

- SiplacePro 10.x
- SiplaceSetupCenter 5.x
- OIB Version 3.x

5.1 Traceability mode

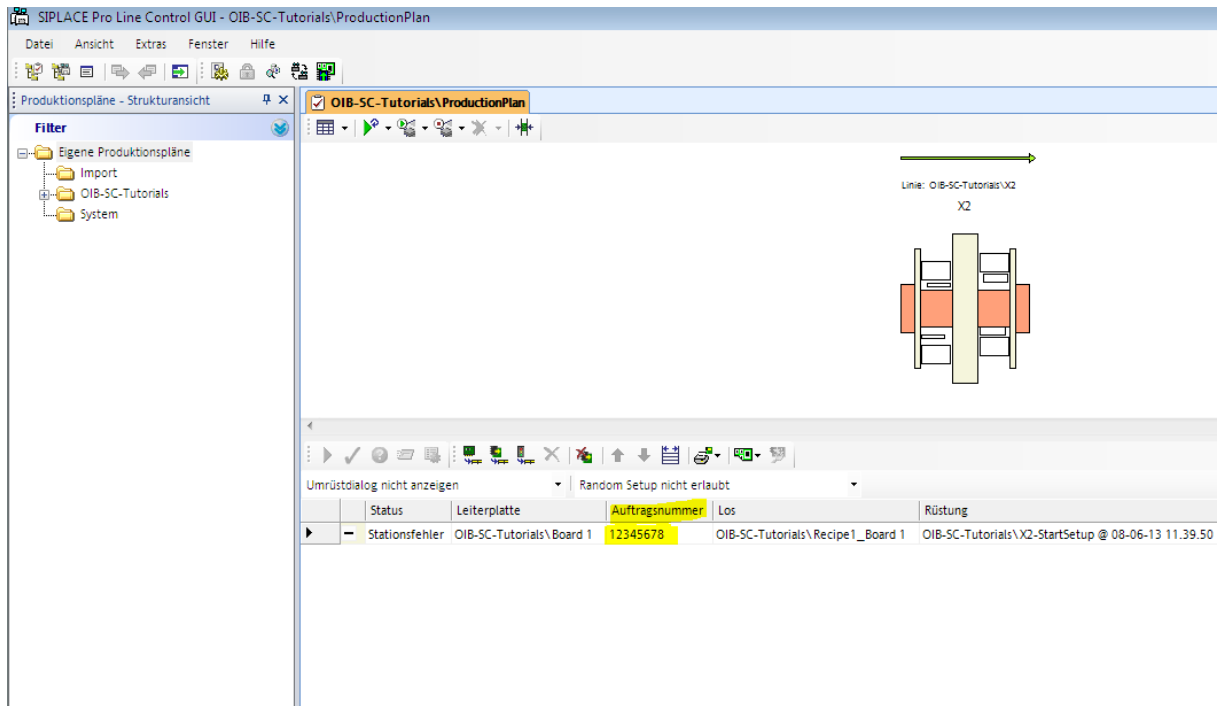
The base to collect traceability data is the mode **LineBasedBoard**. You must set the traceability tool in the Siplace line accordingly.



5.2 MES order number

In order to automatically log on and interrupt operations in HYDRA you need to have MES order numbers located in data supplied by Siplace.

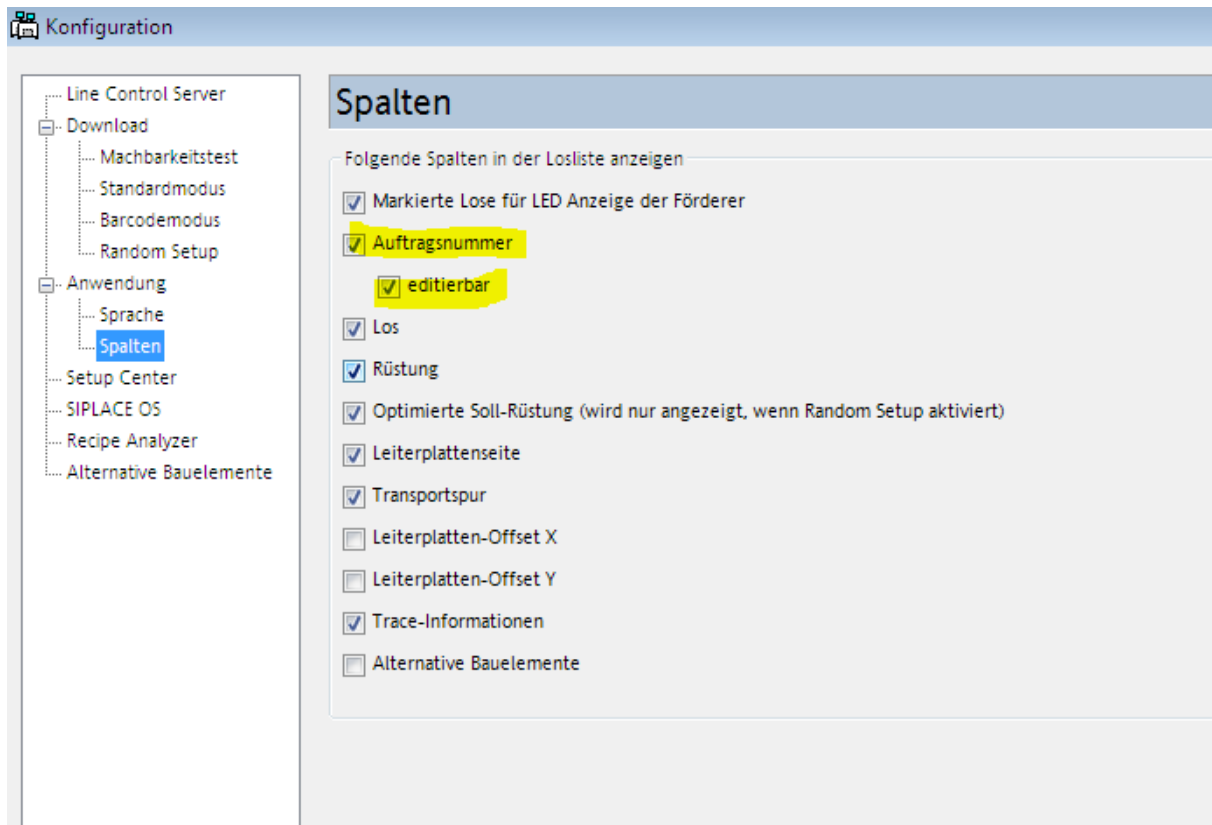
You need to store the MES order number in the SiplaceLineControl before the data record is transferred.



You can switch on the field "Order number" as follows:

Menu: Extras → Configuration

„Application“ → spilt → check „Order number“ and activate „editable“.

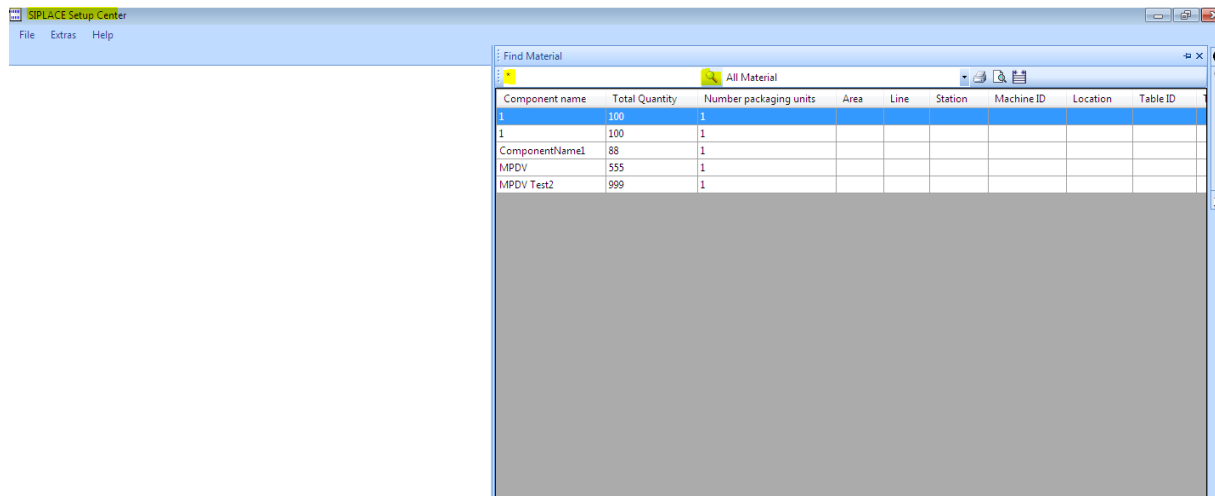


6 Available function tests

6.1 Transfer of batch data list to the SetupCenter of Siplace

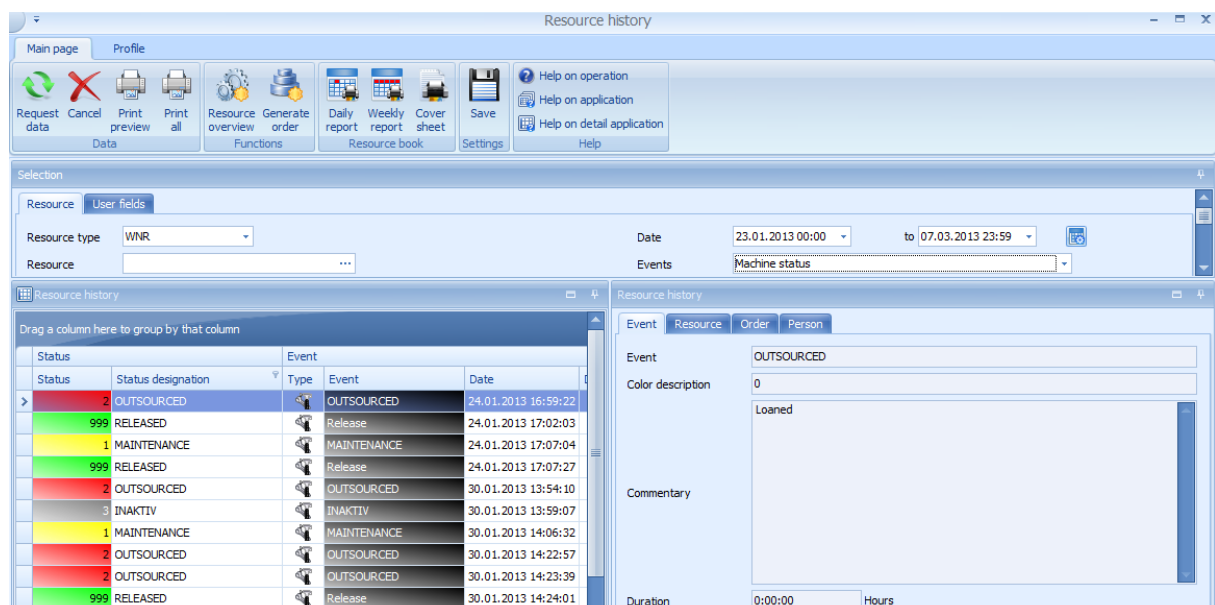
The transferred batches are visible in the SiplaceSetupCenter.

Open "Find material" in the SiplaceSetupCenter and enter "*" as the search criteria. Then all batches should be visible.



6.2 Machine status

You can see if the machine status has been transferred in the resource overview (transaction resov) or in the resource history (transaction reshi). If the status changes for the configured machines appear, the status transfer from Siplace and HYDRA works.



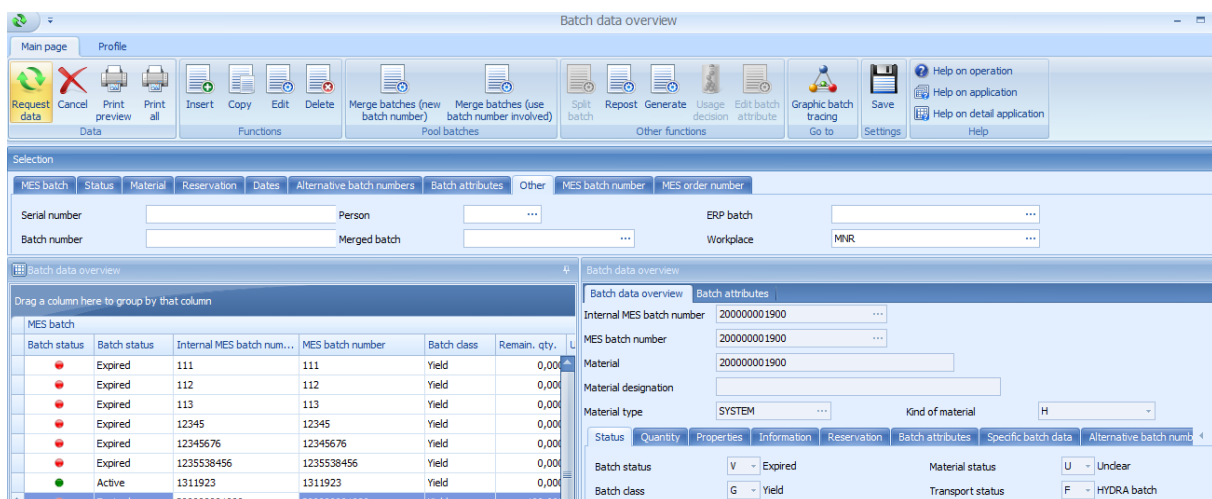
6.3 Order logon / interruption

You can see if orders/operations are logged on or interrupted in the order overview (transaction orov). Select the relevant orders and see if the operation is logged on.

6.4 Create output batches

You can control if output batches (e.g. produced circuit boards) have been created by opening the batch data overview (transaction batov).

Select "Miscellaneous" in the Siplace workstation/machine. Then all produced output batches should be listed.



6.5 Traceability data

You can see in the graphical batch tracing (transaction code battrg) if the assignment of input batches to the output batches had worked.

Select "Origin" and the batch number (if necessary).

Main page Profile

Request data Cancel Print preview Print all Save Help on operation Help on application Help on detail application

Selection

General Attributes

MES batch number Date from 18.02.2007 10:21 to 18.02.2016 23:59

Alternative batch no. 1

☐ Origin ☒ Direct input/output batches

☒ Consumption ☒ Consider long-term data

Graphic batch tracing

MES batch number

- WE13DZX111111118
- PR1NFZX111111118
- PR1NFZX111111117
- PR1NFZX111111116

Batch tracing

Order 55555550200

Workplace 00050500

Short name ROHRMON1

Disposition Dokumentation, Arbeitsblatt 1